

Desmond Kofi Boateng

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RESEARCH PROFILE

PhD student in Computing with a Data Science emphasis and an MSc in Mathematical Sciences, specializing in scientific machine learning, numerical PDEs, and operator learning. Develops Python-based kernel and neural-operator methods and conducts reproducible GPU/HPC experiments across multiple PDE benchmarks. Interested in developing scalable numerical and machine-learning methods for complex scientific and engineering systems governed by partial differential equations.

TECHNICAL SKILLS

- Programming: Python, Julia, Bash; working knowledge of MATLAB and R
- Scientific ML: PyTorch, TensorFlow/Keras, JAX, scikit-learn
- Numerical Computing: NumPy, SciPy, Pandas; numerical PDEs, numerical analysis, kernel methods, numerical linear algebra
- HPC & Development: GPU computing, Slurm, Linux, Git, LaTeX
- Productivity: Microsoft Excel, PowerPoint, Word

EDUCATION

Boise State University <i>PhD in Computing, Data Science Emphasis</i> GPA: 3.85/4.00 Advisor: Professor Grady Wright Research focus: scientific machine learning, numerical PDEs, and operator learning	Expected Dec. 2028 Boise, ID
African Institute for Mathematical Sciences <i>MSc in Mathematical Sciences</i> Thesis: Comparative Analysis of Singular Value Decomposition Based Methods for Image and Video Compression with Julia	Jun. 2024 Kigali, Rwanda
University of Ghana <i>BA in Mathematics with Economics</i> Thesis: Khovanov's Categorification of Jones' Polynomial	Nov. 2021 Accra, Ghana

RESEARCH EXPERIENCE

Graduate Research Assistant <i>Boise State University</i>	Aug. 2025 – Present Boise, ID
• Develop and benchmark kernel-based and neural-operator methods for learning solution operators arising from PDEs. • Implement reproducible scientific-machine-learning pipelines in Python using PyTorch, TensorFlow, JAX, NumPy, and SciPy across PDE benchmarks including advection, Burgers' equation, Darcy flow, Helmholtz problems, structural mechanics, and Navier–Stokes. • Design and execute GPU-accelerated experiments and Slurm workloads on Boise State University's Borah HPC cluster, including hyperparameter sweeps, model comparisons, and reproducible experiment tracking. • Developed a more computationally efficient approach for evaluating the W2 transform in graph-wavelet analysis.	

RESEARCH OUTPUTS AND PRESENTATIONS

- Accepted poster: Boateng, D. K., and G. B. Wright. "Cross-Grid Generalization for Some Operator Learning Frameworks." SIAM Conference on Mathematics of Data Science, Salt Lake City, Utah, Nov. 16–20, 2026.
- Manuscript in preparation: Boateng, D. K., and G. B. Wright. "Adaptive Kernel Expansion DeepONets (AKED)."
- Poster: "Polyharmonic Splines for Operator Learning." CBMS-AMML, University of Houston, Dec. 2025; SIAM Pacific Northwest Biennial Meeting, University of Washington, Oct. 2025.
- Talk: "Singular Value Decomposition Based Methods for Image and Video Compression with Julia." Conference on Mathematics and Its Applications, AIMS Ghana, 2023.

RELEVANT PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Feb. 2024 – Jul. 2024

Cape Wesley Consulting

Remote

- Built a LegalBERT-based classifier to identify obligations in the U.S. Code of Federal Regulations, achieving 95% accuracy on the evaluation set.
- Developed Python pipelines to collect, extract, and process XML-based regulatory text for downstream policy analysis.

Graduate Teaching Assistant

Aug. 2024 – May 2025

Boise State University

Boise, ID

- Led tutorials for MATH 307: Foundations of Cryptography and modernized a nine-year-old course codebase.
- Developed grading rubrics and assessed work for MATH 475/575, communicating technical feedback to students.

SELECTED LEADERSHIP AND SERVICE

- President, International Student Association, Boise State University (2025–2026): Led student programming and collaborated with university stakeholders.
- Graduate College Senator, ASBSU (2025–2026): Represented graduate student interests in university discussions and initiatives.
- Programs Manager, MeetMentors (Apr. 2021–Present): Developed the Xplore program serving approximately 3,000 university students and led EQUIP programming reaching approximately 200 young people across Africa.

SELECTED HONORS AND PROFESSIONAL DEVELOPMENT

- Selected participant, Kernel Approximation and Gaussian Processes: Integrating and Expanding Perspectives, Banff International Research Station, Sept. 13–18, 2026.
- Selected participant and \$1,300 travel-grant recipient, US Research Software Sustainability Institute Summer School, Northeastern University, Jun. 2026.
- Conference travel grants: CBMS-AMML (\$900, 2025) and SIAM Pacific Northwest Biennial Meeting (\$500, 2025).